SAheart: support vector classification

Data and procedure

Variables: sbp, typea, age (integer); tobacco, ldl, adiposity, obesity, alcohol (real); famhist (string category); chd **(outcome)** (integer). All predictors standardized. 462 rows; CHD 0/1 counts 302/160.

Missing value: no (no NA/blank/sentinel codes observed); no duplicates. **Recoding:** famhist Absent=0, Present=1 (462 values); chd unchanged, 0=negative, 1=positive. All rows and IQR-flagged extremes retained by user choice; no imputation, trimming or other source-data changes.

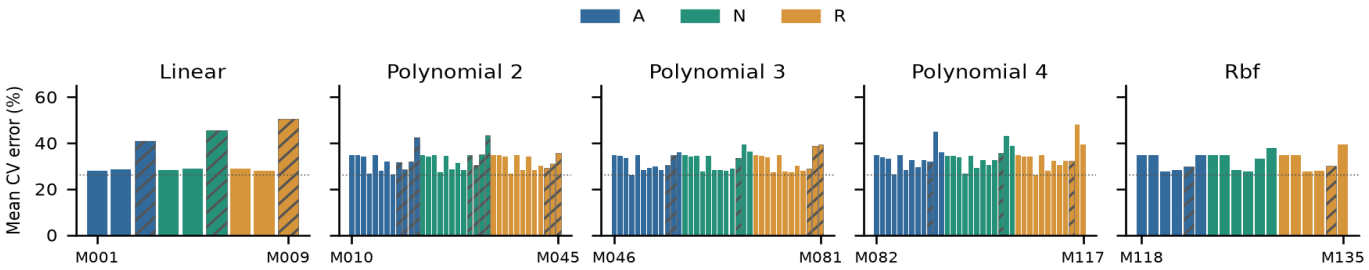
CV: Python/scikit-learn 1.9.1, train_test_split(stratify=chd), seed 43209: 369 train / 93 test. StratifiedKFold(10, shuffle=True, same seed), shared folds. Each fold fits scales and squared-term scales on its training rows, predicts validation labels, then computes accuracy=correct/n, error=incorrect/n and TN/FP/FN/TP. SE=sample fold SD/sqrt(10).

Compared models: A=all 9; N=A + squared standardized age/tobacco (11); R=A - alcohol (8). Training quartile CHD rates suggest saturation for age/tobacco; alcohol has weak marginal association (r=0.065). These are potential nonlinear/redundant predictors, not established findings.

Kernel	Representations / degree	C	gamma	coef0
Linear	A, N, R / NA	0.1, 1, 10000	NA	NA
Polynomial	A, N, R / 2, 3, 4	0.1, 1, 10000	0.01, 0.1	0, 1
Radial (RBF)	A, N, R / NA	0.1, 1, 10000	0.01, 0.1	NA

Full Cartesian grids (135 configurations): C spans regularization through the required high-C hard-margin approximation; gamma spans smooth/local behavior; coef0 contrasts homogeneous/shifted polynomials. No extra classifiers. SVC tol=0.001, max_iter=200000.

Results



Every configuration shown in ID order: blue A, green N, gold R; hatching = 25 candidates with nonconverged folds (208/1350 fits; provisional, excluded from ranking). Exact IDs, grids, fold results and warnings: analysis/cv_models.csv and cv_folds.csv.

Model (kernel / representation / parameters)	CV error % +/- SE	TN +/- SE	FP +/- SE	FN +/- SE	TP +/- SE
M109 poly4/R; C=0.1, g=0.1, c0=1	25.99 +/- 1.96	22.00 +/- 0.42	2.10 +/- 0.38	7.50 +/- 0.67	5.30 +/- 0.60
M049 poly3/A; C=0.1, g=0.1, c0=1	26.27 +/- 1.59	22.00 +/- 0.39	2.10 +/- 0.38	7.60 +/- 0.65	5.20 +/- 0.57
M017 poly2/A; C=1, g=0.1, c0=1	26.53 +/- 1.85	21.10 +/- 0.53	3.00 +/- 0.52	6.80 +/- 0.61	6.00 +/- 0.52
M085 poly4/A; C=0.1, g=0.1, c0=1	26.53 +/- 1.62	21.80 +/- 0.44	2.30 +/- 0.42	7.50 +/- 0.48	5.30 +/- 0.40
M097 poly4/N; C=0.1, g=0.1, c0=1	26.80 +/- 1.88	21.70 +/- 0.40	2.40 +/- 0.37	7.50 +/- 0.62	5.30 +/- 0.52

Confusion cells are decimal mean validation-fold counts +/- SE; no training metrics. Top five converged configurations shown; all other close/tied candidates remain in analysis/cv_ranked.csv.

Selected model and estimated prediction error

1. Chosen model: M049, poly degree 3, representation A, 9 dimensions; C=0.1, gamma=0.1, coef0=1.

2. Refitting and application: Refit preprocessing and classifier on all 369 training rows, then predict the 93 reserved test rows once.

3. Testing performance: error 30.11% (28/93); accuracy 69.89% (65/93).

4. Testing confusion matrix and error types: negative=0, positive=1; **TN=52, FP=9, FN=19, TP=13**. Type I=FP/(TN+FP)=9/61=14.75%; Type II=FN/(FN+TP)=19/32=59.38%.

Main findings and limitations

Best CV errors by representation: A 26.27%, N 26.80%, R 25.99%; reduction helps marginally, extra terms do not. With user discretion, M049 is preferred for lower degree and lower fold SE than M109. The 462-row sample is imbalanced (34.6% positive); compare class errors above. Feature nomination before CV and extensive tuning cause selection optimism; fold SEs ignore fold dependence and are descriptive. Small differences do not prove superiority. No external validation; generalization and stability remain uncertain. High-C failures prevent a general hard-margin conclusion.

Reproduction, data changes, distributions, outliers, complete grids, convergence diagnostics, selection and code: analysis/README.md; analyze_saheart.py; report_saheart.py.

Appendix: SKILL.md

Full analysis instructions, reproduced from the local skill file.

---

name: lda-qda-analysis

description: Fit SVM a one-page report.

General Rules

- Linear model with standardized predictors (and no extra predictors) should always be included.

- Report every data change, reason, affected variables/counts and recoding mappings/levels. Keep a reproducible record; summarize changes in report. State when none.

- Report <=1 A4 page total. Render/check pagination; Markdown alone cannot verify it. Reference code/diagnostics/overflow tables in analysis artifact. Disclose unavailable results, failures or incomplete CV.

- When testing penalization parameter C, C = 10000 should be tested for every model to see if hard margin works.

Steps of Analysis

1. Preprocessing

- Identify outcome/predictors; ask if unclear. Inspect dimensions, types, rows, summaries, classes, distributions, outliers, duplicates and missingness counts/patterns/codes.

- If missingness exists, ask the user if at least missing at random is a reasonable assumption.

- If a variable appears like categorical variable, ask user if it should be treated as numerical or categorical. Standardize each predictor. Predictors mentioned later are all referring to standardized predictors unless further specified.

- When you are suspicious of mistakes, such as outliers or duplicated data, ask the user for how that want to be dealt with.

- Record all the changes you made to the dataset. All modification needs to be reported in the final report.

2. Exploration

- Check the distribution of variables, if there are clear non-linear relationship between a predictor and the outcome (e.g., sinx), remember them. Call these predictors potential non-linear relationship predictors.

- Check for clearly unrelated predictors. Remember them. Called these predictors potential redundant predictors.

3. Fir and Compare models

- Reserve test data before fitting; honor existing splits. Ask reader which of 5, 10 or 20 folds is preferred, or if they are willing to let you decide. Record seed, package, function, and language you used for splitting. Same CV folds should be used for all models.

- Fit models with linear kernel with original predictors, original + potentiall non-linear relationship predictors, original - potential redundant predictors. Test different penalization parameters. You will decide the penalization parameters you try, but you have to record them and explain in the report.

- Fit models with polynomial kernel with original predictors, original + potentiall non-linear relationship predictors, original - potential redundant predictors. Test different penalization, gamma, and coef0 parameters. You will decide all the penalization you try, but you have to record them and explain in the report. This should be done for all degree = 2, 3, and 4.

- Fit models with radial bases kernel with original predictors, original + potentiall non-linear relationship predictors, original - potential redundant predictors. Test different penalization and gamma parameters. You will decide all the penalization you try, but you have to record them and explain in the report.

- The above models are required. Feel free to fit other models you believe to have good potential. For these models, you decide the parameters you try out. Remember these extra models and explain them in final report.

- Remember the at least 3 models with the lowest CV mean misclassification rate and their respective standard error for CV misclassification rate. If 4th or 5th model is also close competitive, remember these result from them as well.

- If the model with the lowest CV mean misclassification rate is not the one with the lowest standard error for CV misclassification rate among the models you remembered, ask user how to resolve this, also ask them if they wish to let you decide.

Report

1. Data and procedure: list with these headings

- **Variables:** Original predictor names/types; outcome name followed by **(outcome)** and original type. Remind reader that you standardized.

- **Missing value: no**, or briefly describe affected variables and treatment.

- **Recoding:** Only recoded variables and mappings, including outcome levels.

- **CV:** Seed, training/testing sizes, fold count, shared folds, fold-fitted preprocessing and each iteration's accuracy/error/matrix calculation. No fold-count rationale.

- **Compared models:** Define what is your potential redundant predictors and potential non-linear relationship predictors and use them in report to save space. Briefly describe every candidate, including the predictors, degree, gamma, coef0 and the penalization parameter you tried. If possible, organize this in a table and put NA in cells that do not apply (such as coef0 for radial kernel)

Results

Figure: It should be a bar chart presenting mean CV misclassification rate for every model you tried. You decide if this should be horizontal or vertical. Make it easy to read but don't occupy too much space.

Table: The top performing models (i.e., the 3 models with the lowest mean CV misclassification rate and their close competitors), no training metrics/matrices. Binary confusion cells: TN, FP, FN, TP as decimal mean fold counts with SE = sample SD of fold counts / sqrt(k).

Selected model and estimated prediction error: numbered list

1. **Chosen model:** Classifier and representation/dimension; no repeated CV error.

2. **Refitting and application:** Refit preprocessing/model on whole training set, then apply once to reserved testing set; no training performance.

3. **Testing performance:** Error and accuracy percentages with incorrect/total and correct/total counts.

4. **Testing confusion matrix and error types:** State negative/positive classes; explicitly label integer counts in TN, FP, FN, TP order. Type I (false positive) = FP/(TN+FP); Type II (false negative) = FN/(FN+TP). Give percentages and counts; zero denominator means undefined. Multiclass: full labeled matrix and per-class one-versus-rest rates. If testing labels unavailable, mark metrics unavailable, not CV-as-test.

Main findings and limitations

Explain preference, reduction effects, class errors and limitations: size, imbalance, assumptions, stability/generalization. Note selection optimism; small differences do not prove superiority.